

ThreadLine

Apparel & Merchandising Checkout System

Complete Operational Workflow Guide

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Prepared for: Store operators, merchandising teams, and stakeholders

Version: 1.0 - Apparel retail edition

1. Executive Summary

ThreadLine is a self-checkout and store-operations platform built specifically for clothing shops, boutiques, and merchandising retailers. It replaces generic grocery-style POS with workflows that understand size/color variants, seasonal collections, hang-tag barcodes, and the high return rates typical of fashion retail.

The system connects four touchpoints - customer self-checkout, admin merchandising HQ, cashier counter settlement, and exit-gate verification - through a single API and inventory ledger. Every sale, refund, and stock adjustment is recorded in real time.

2. Industry Gaps in Apparel Retail (2025-2026)

Research across apparel POS guides, inventory management platforms, and fashion retail operations identifies these critical gaps:

Gap	Why it matters
Variant-heavy inventory	One style x multiple sizes and colors creates thousands of SKUs. Flat product ca
Matrix stock visibility	Selling out of M/Black while L/Navy sits on shelf is invisible without per-varia
Returns & exchanges	25-40% return rates in apparel require fast size swaps, not full-order refunds o
Seasonal markdowns	Collections turn every 12-16 weeks; manual price changes cause errors.
Omnichannel drift	Online and in-store stock must share one ledger to prevent overselling.
Self-checkout for fashion	Customers expect to scan hang-tags, filter by size, and pay at counter or online
Low-stock per variant	Reorder triggers must fire at size/color level, not style level.
Merchandising analytics	Reports need breakdown by department, style, color, and size - not just revenue.

3. How ThreadLine Addresses These Gaps

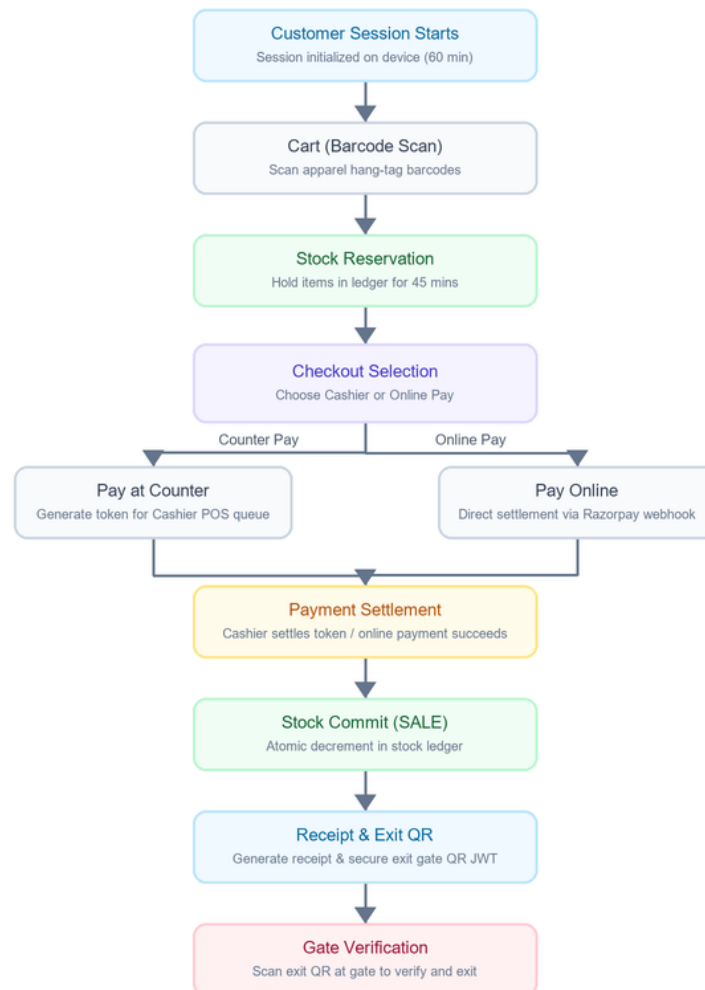
Capability	ThreadLine approach
Style-color-size SKU model	Each variant is a product row with style code, size, color, brand, season, auto-
Variant inventory matrix	Admin filters by size and color; low-stock alerts fire below 5 units per variant
Stock ledger & reservations	Cart holds and unpaid orders reserve stock for 45 minutes; sales commit atomical
Self-checkout + counter pay	Customer builds bag on phone; pays at counter with token or online via Razorpay.
Promotions per SKU	Percentage discounts on individual variants for clearance and seasonal markdowns
Returns with restock	Manager refunds paid orders and restores inventory through the ledger.
Exit gate security	One-time QR pass prevents walkouts without settlement.
Merchandising dashboard	Overview shows revenue, margin, category mix, top demand styles, and stock healt

4. System Architecture

ThreadLine is a TypeScript monorepo with four applications and one API:

- Customer Shop (port 3001) - Browse, scan hang-tags, filter by category/size, checkout
- Admin HQ (port 3002) - Catalogue variants, stock, promotions, orders, audit log
- Cashier Terminal (port 3003) - Settle counter orders, exit gate verification
- API Service (port 4000) - Express + PostgreSQL (Supabase), JWT auth, stock ledger
- Database - products, stock_movements, stock_reservations, orders, receipts, audit_log

Data Flow Diagram



5. Roles & Access

Role	Access
Customer	Self-checkout shop, session recovery by phone, receipt download
Staff	Add variants, adjust stock, view orders, set promotions, CSV export
Manager	Void unpaid orders, refund paid orders, audit log, order CSV export
Cashier	Settle counter tokens, view pending queue, verify exit passes

6. Merchandising Workflow - Setting Up the Catalogue

Step 1: Define the style

For each design (e.g. Slim Fit Oxford Shirt), choose a style code (OXF-2026), brand, category (Men's Wear, Women's Wear, etc.), season/collection (SS26, AW26, Core, Clearance), and gender.

Step 2: Create one SKU per size/color

Each size/color combination gets its own row: unique hang-tag barcode, size (S/M/L/XL or numeric), color, cost price, list price, tax %, opening stock, and product image. SKU auto-generates as STYLE-COLOR-SIZE (e.g. OXF-2026-NAVY-M).

Step 3: Receive stock

When shipment arrives, use Set Stock in Admin to update quantity. Every change writes to the stock ledger (INITIAL, ADJUSTMENT, CORRECTION) with actor and note for audit.

Step 4: Monitor variant health

Filter inventory by size or color. Low-stock view highlights variants below 5 units. Reserved column shows units held in active carts or unpaid orders.

Step 5: Run promotions

Apply percentage discounts to specific variants for end-of-season clearance. Effective shelf price updates instantly on the customer shop.

7. Customer Self-Checkout Workflow

- Step 1: Customer opens shop URL (?store=BLR001 for multi-store). Session auto-starts (60 min).
- Step 2: Optional: enter phone number to recover cart if they leave and return.
- Step 3: Browse by category (Men's Wear, Footwear, Accessories, etc.) or filter by size.
- Step 4: Scan hang-tag barcode with camera or search by name, style, color, SKU.
- Step 5: Add items to bag; adjust quantities. Out-of-stock variants show ribbon.
- Step 6: Proceed to checkout: choose Pay at Counter (token number) or Pay Online (Razorpay).
- Step 7: For counter: receive token number; proceed to cashier with items.
- Step 8: For online: complete payment; system marks order paid on webhook.
- Step 9: Download receipt; show exit QR pass at store gate.

8. Cashier & Counter Settlement Workflow

The cashier terminal is optimized for fashion retail counter service:

- Pending queue shows all unpaid counter orders with token numbers.
- Lookup by token number or order UUID.
- Review line items, totals, and customer bag.
- Settle - marks paid, commits stock (SALE in ledger), issues receipt.
- Customer receives receipt and exit pass.

Exit gate

Security scans the one-time exit QR JWT. System verifies payment, prevents replay, and logs gate passage. Anti-walkout control for high-shrink apparel categories.

9. Returns & Refunds Workflow

Fashion retail returns are handled at manager level:

- Locate order in Admin > Orders by receipt or order ID.
- Manager issues refund on paid order - stock restored via REFUND ledger entry.
- Void available for unpaid/abandoned counter orders - releases reservation.
- Resend receipt email if customer lost original.

Roadmap: size-for-size exchanges without full refund + new sale (common apparel gap).

10. Stock Ledger & Reservations

Event	Ledger reason
New variant created with stock	INITIAL
Customer adds to cart	Reservation (session hold)
Checkout without payment	Reservation (order hold, 45 min)
Payment settled	SALE (stock decremented)
Refund processed	REFUND (stock restored)
Manual count correction	ADJUSTMENT or CORRECTION

11. Reporting & Analytics (Admin Overview)

- Revenue, paid vs open orders, average order value
- Inventory value at cost and at list price
- Low-stock variant count and reserved units
- Category breakdown - units and value by department
- Top demand styles (demand score driven featuring)
- Margin per unit at list and promotional price
- CSV export: full variant catalogue and order history (manager)

12. Apparel Categories Supported

- Men's Wear, Women's Wear, Kids & Baby, Unisex
- Footwear, Accessories, Bags & Luggage
- Activewear, Ethnic & Traditional
- Innerwear & Loungewear, Winterwear
- Merchandise & Gifts

13. Summary

ThreadLine transforms a generic checkout engine into an apparel-aware retail platform. By treating every size/color as a tracked SKU with style grouping, real-time reservations, and variant-level low-stock alerts, it closes the biggest operational gaps that clothing and merchandising shops face when adopting self-checkout.

This document describes the complete workflow from catalogue setup through customer checkout, counter settlement, stock ledger management, and returns - ready for store teams and stakeholders.